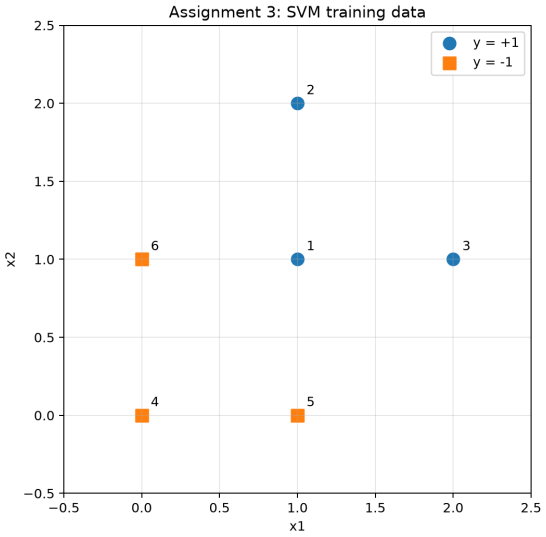
Assignment 3: SVMs, K-means, and K-NN

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Question 1: Hard-margin linear SVM

(a) Plot and support vectors



Support vectors: **points 1** $(1,1)$**, 5** $(1,0)$**, and 6** $(0,1)$**.**

The classes are separated by lines of slope -1. Point 1 lies on $x_1 + x_2 = 2$ and points 5 and 6 lie on $x_1 + x_2 = 1$; every other point is farther out (points 2 and 3 on $x_1 + x_2 = 3$, point 4 on $x_1 + x_2 = 0$). The widest band between the classes is therefore $1 \le x_1 + x_2 \le 2$, and its two edges touch exactly points 1, 5, and 6. These are the closest opposite-class points, so they determine the boundary.

(b) Boundary and margin

Each support vector lies on its supporting line, $w \cdot x + b = y$:

$$w_1 + w_2 + b = +1 \quad \text{(point 1)}, \qquad w_1 + b = -1 \quad \text{(point 5)}, \qquad w_2 + b = -1 \quad \text{(point 6)}.$$

The last two give $w_1 = w_2$. Subtracting the second from the first gives $w_2 = 2$, so $w_1 = w_2 = 2$ and $b = -3$.

Decision boundary: $2x_1 + 2x_2 - 3 = 0 \quad (w = (2,2),\ b = -3)$.

Check: $y_i(w \cdot x_i + b)$ equals $1, 3, 3, 3, 1, 1$ for points 1–6, so every point satisfies $y_i(w \cdot x_i + b) \ge 1$, with equality only at the support vectors.

Geometric margin (distance from the boundary to the closest training points):

$$\gamma = \frac{1}{\|w\|} = \frac{1}{\sqrt{2^2 + 2^2}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4} \approx 0.354.$$

Full width between the supporting lines (reported separately): $2/\|w\| = \sqrt{2}/2 \approx 0.707$.

Question 2: K-means

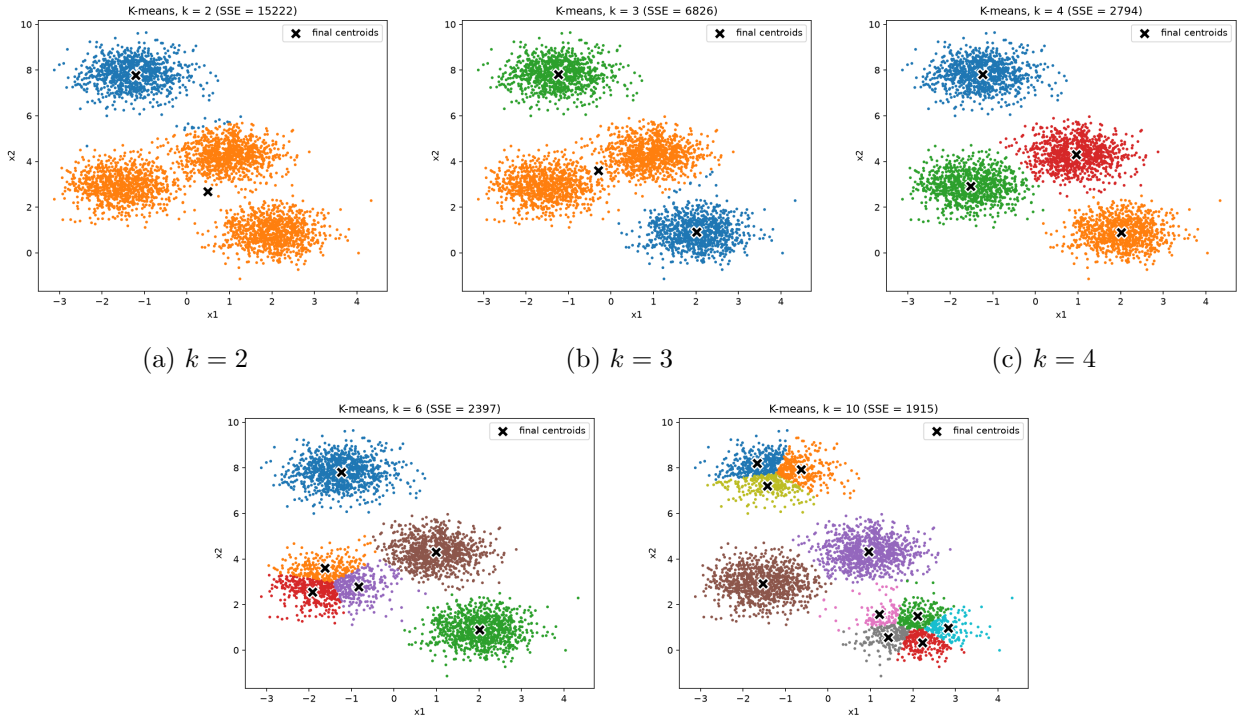
(a) Implementation notes

Distance ties: a point equally close to several centroids is assigned to the centroid with the lowest index (`np.argmin` returns the first minimum).

Empty clusters: a cluster that receives no points keeps its previous centroid.

(b) Results for $k = 2, 3, 4, 6, 10$

Run with `-seed 0 -max-iterations 100`.



(a) $k = 2$ (b) $k = 3$ (c) $k = 4$

(d) $k = 6$ (e) $k = 10$

Figure 1: Cluster assignments and final centroids (black X).

With $k = 2$ and $k = 3$ there are fewer clusters than generating groups, so K-means merges whole groups: three groups share one cluster at $k = 2$, and the two closest groups share one at $k = 3$. At $k = 4$ each cluster matches one of the four generating groups (about 1,000 points each). At $k = 6$ and $k = 10$ the extra centroids split real groups into smaller pieces that do not correspond to any structure in the data. The within-cluster sum of squares keeps falling as k grows, but most of the drop happens by $k = 4$.

Question 3: Noisy-label SVM and K-NN repair

Run with `-repeats 20 -seed 0`. Per-trial results are in `results/noise_comparison.csv`.

Method	Mean clean test accuracy (20 trials)
(a) Noisy-label baseline	0.8280
(b) K-NN label repair	0.8644
Difference (K-NN − baseline)	+0.0364

Did K-NN repair help? Yes. It improved accuracy by about 3.6 percentage points on average and was better in all 20 trials. Repair helps because flipped labels deep inside a class are outvoted by their correctly labeled neighbors and corrected; those points otherwise pull the SVM boundary out of place. It also offsets the asymmetric noise: positives are flipped more often (0.35 vs. 0.20), which pushes the baseline boundary into the positive class. Repair can hurt where the classes overlap, since there the majority vote can relabel correct minority points, and in any neighborhood where more than half the labels are wrong, since the vote then spreads the error.

Acknowledgment of outside help

I used a generative AI assistant (Claude, Anthropic, via Claude Code) on this assignment. It helped with the derivations, the code in the three template files, the experiments, the figures and the formatting of this report.